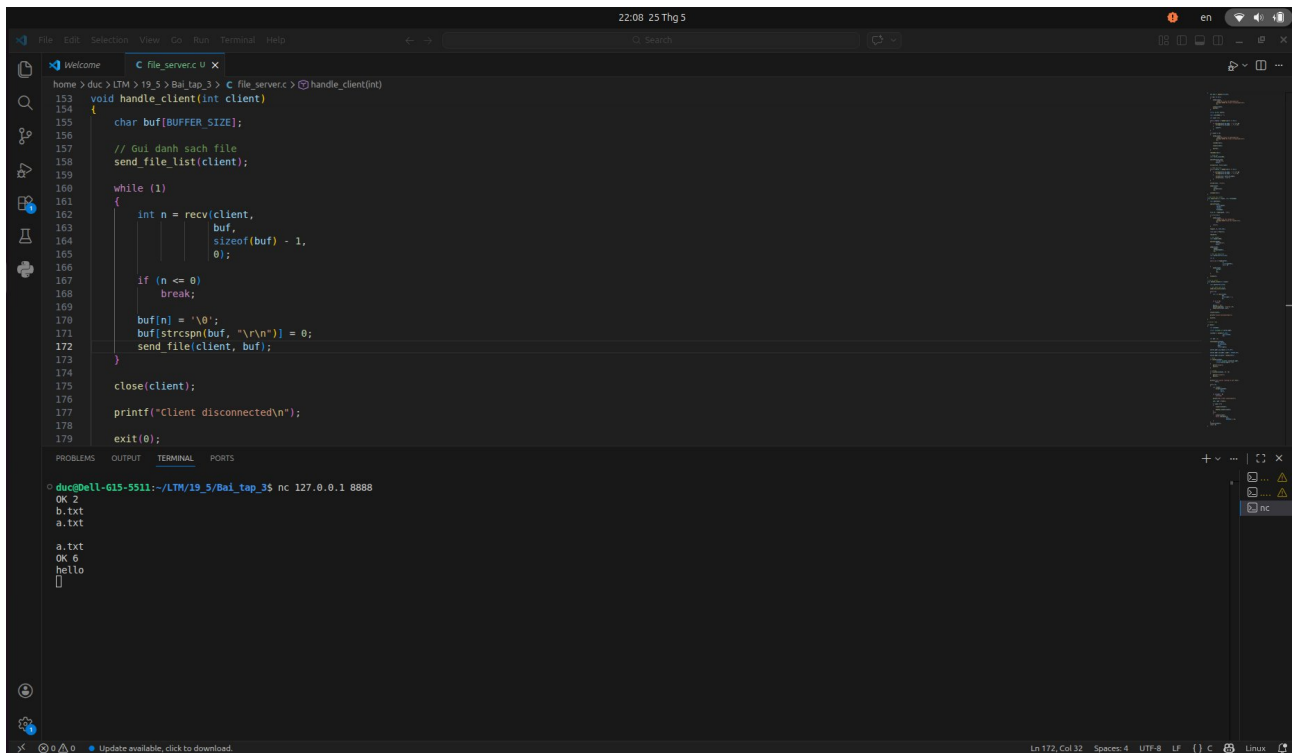


Bùi Anh Đức-20225285

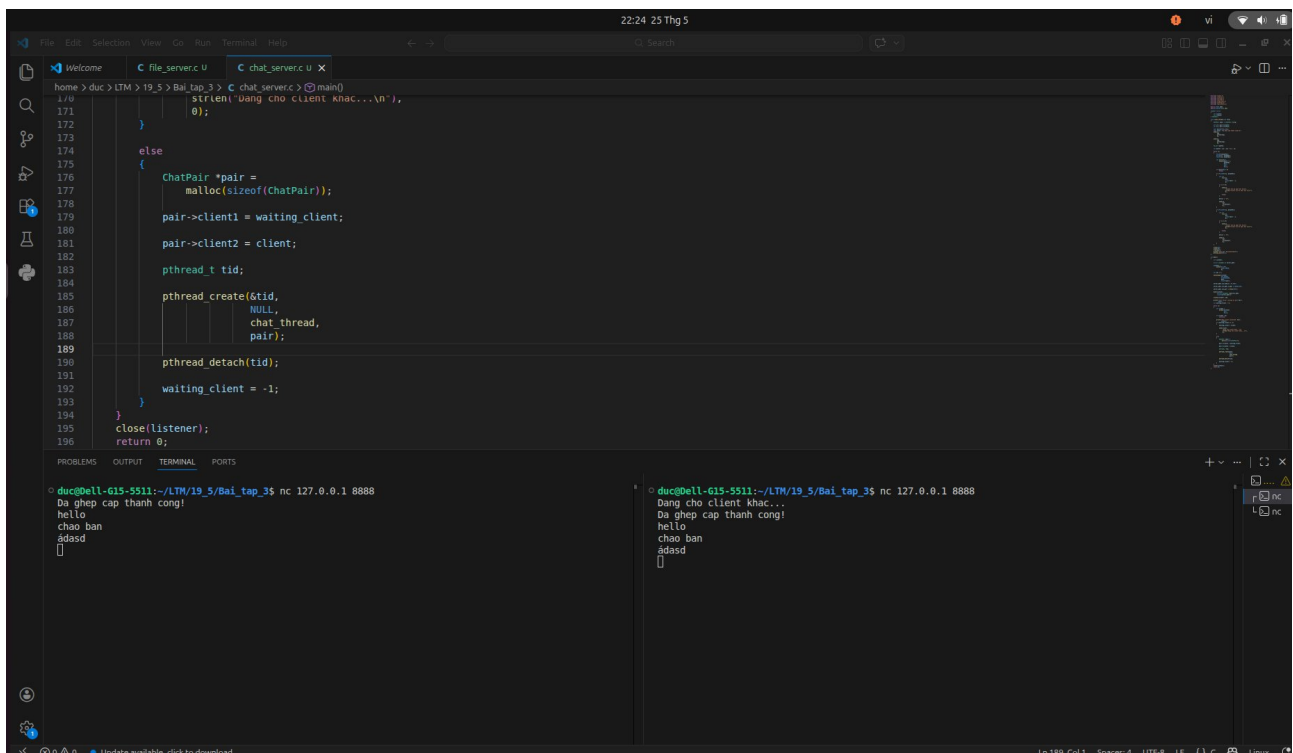
Bài tập 19/5

Bài 3.1



The screenshot shows a VS Code editor with a C program named `file_server.c` and its terminal output. The program is a simple file server that listens on port 8888. It receives a list of files from a client and then sends them back. The terminal output shows the server's response to a client connection.

```
home > duc > LTM > 19.5 > Bai_tap_3 > C file_server.c > handle_client(int)
153 void handle_client(int client)
154 {
155     char buf[BUFFER_SIZE];
156     // Gai danh sach file
157     send_file_list(client);
158     while (1)
159     {
160         int n = recv(client,
161             buf,
162             sizeof(buf) - 1,
163             0);
164         if (n <= 0)
165             break;
166         buf[n] = '\0';
167         buf[strlen(buf) - 1] = '\n';
168         send_file(client, buf);
169     }
170     close(client);
171     printf("Client disconnected\n");
172     exit(0);
173 }
174
175 PROBLEMS OUTPUT TERMINAL PORTS
+ ~ ~ ~ ~ ~
o duc@Dell-615-5511:~/LTM/19_5/Bai_tap_3$ nc 127.0.0.1 8888
OK 2
b.txt
a.txt
a.txt
OK 6
hello
[]
```

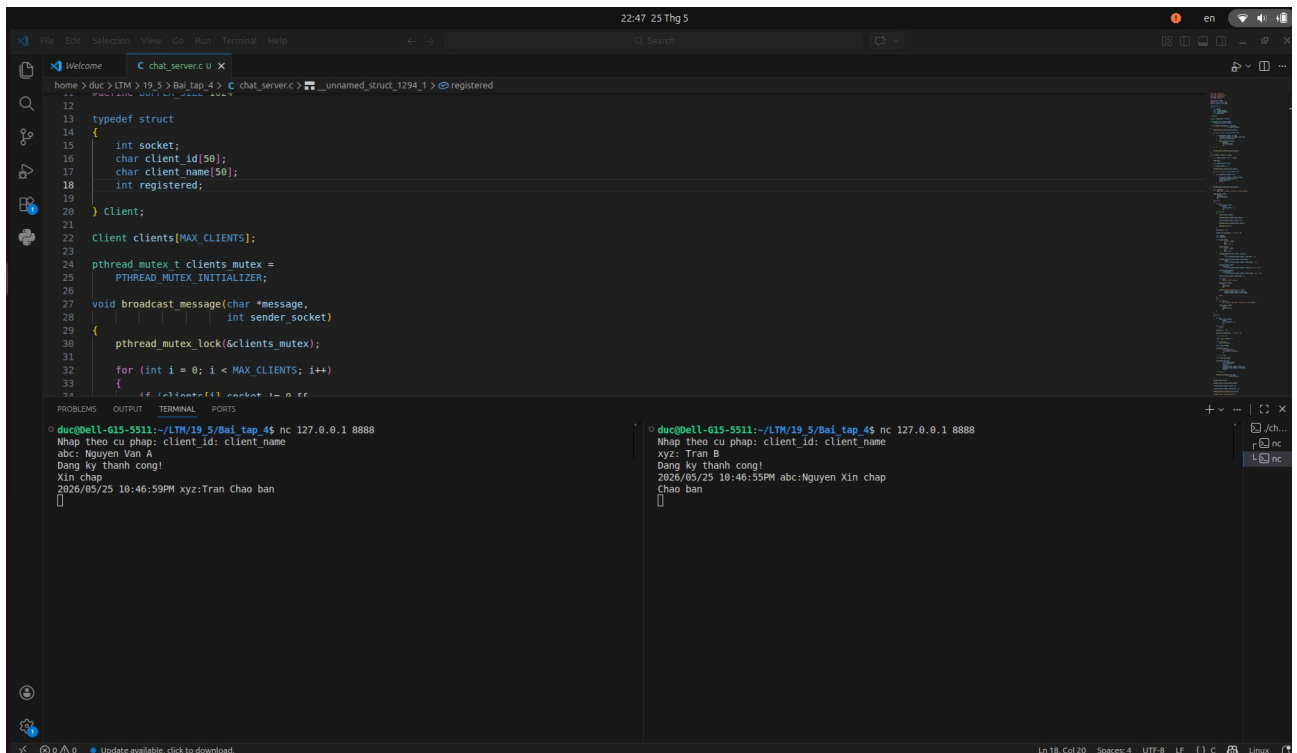


The screenshot shows a VS Code editor with a C program named `chat_server.c` and its terminal output. The program is a chat server that listens on port 8888. It receives a list of files from a client and then sends them back. The terminal output shows the server's response to a client connection.

```
home > duc > LTM > 19.5 > Bai_tap_3 > C chat_server.c > main()
171 //
172     strlen("Dang cho client khac...\n");
173     }
174     else
175     {
176         ChatPair *pair =
177             malloc(sizeof(ChatPair));
178         pair->client1 = waiting_client;
179         pair->client2 = client;
180         pthread_t tid;
181         pthread_create(&tid,
182             NULL,
183             chat_thread,
184             pair);
185         pthread_detach(tid);
186         waiting_client = -1;
187     }
188     close(listener);
189     return 0;
190 }
191
192 PROBLEMS OUTPUT TERMINAL PORTS
+ ~ ~ ~ ~ ~
o duc@Dell-615-5511:~/LTM/19_5/Bai_tap_3$ nc 127.0.0.1 8888
Da ghep cap thanh cong!
hello
chao ban
adasd
[]
o duc@Dell-615-5511:~/LTM/19_5/Bai_tap_3$ nc 127.0.0.1 8888
Dang cho client khac...
Da ghep cap thanh cong!
hello
chao ban
adasd
[]
```

Bài 3.2

Bài 4.1 chat_server



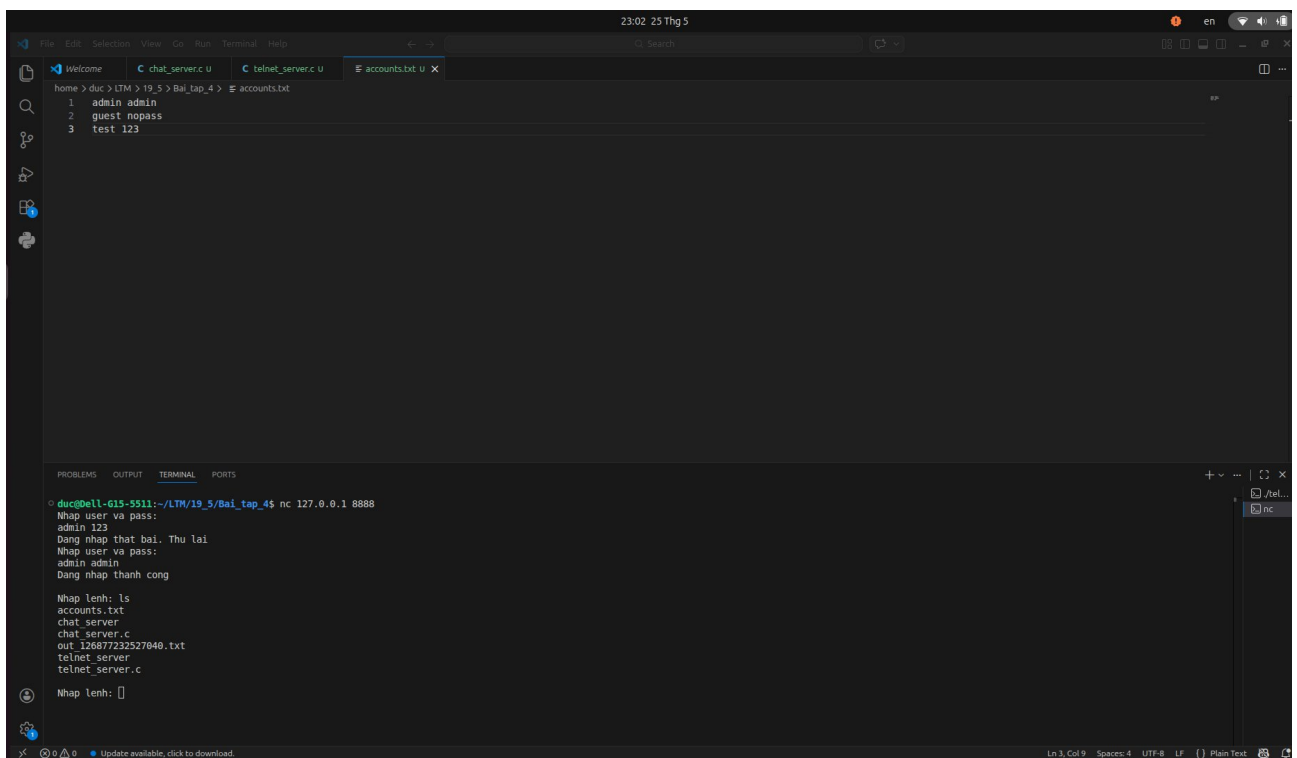
```
22:47 25 Thg 5
File Edit Selection View Go Run Terminal Help
C chat_server.c U X
home > duc > LTM > 19_5 > Bai_tap_4 > C chat_server.c > __unnamed_struct_1294_1 > registered
12
13 typedef struct
14 {
15     int socket;
16     char client_id[50];
17     char client_name[50];
18     int registered;
19 } Client;
20
21
22 Client clients[MAX_CLIENTS];
23
24 pthread_mutex_t clients_mutex =
25     PTHREAD_MUTEX_INITIALIZER;
26
27 void broadcast_message(char *message,
28                        int sender_socket)
29 {
30     pthread_mutex_lock(&clients_mutex);
31     for (int i = 0; i < MAX_CLIENTS; i++)
32     {
33         if (clients[i].socket != 0 &&
34             clients[i].registered)
35         {
36             send(clients[i].socket, message, strlen(message), 0);
37         }
38     }
39 }
40
41 int main()
42 {
43     int server_socket = socket(AF_INET, SOCK_STREAM, 0);
44     if (server_socket == -1)
45     {
46         perror("socket");
47         return 1;
48     }
49     struct sockaddr_in server_addr;
50     server_addr.sin_family = AF_INET;
51     server_addr.sin_port = htons(8888);
52     inet_pton(AF_INET, "127.0.0.1", &server_addr.sin_addr);
53     bind(server_socket, (struct sockaddr *)&server_addr, sizeof(server_addr));
54     listen(server_socket, 5);
55     while (1)
56     {
57         struct sockaddr_in client_addr;
58         socklen_t client_addr_len = sizeof(client_addr);
59         int client_socket = accept(server_socket, (struct sockaddr *)&client_addr, &client_addr_len);
60         if (client_socket == -1)
61         {
62             perror("accept");
63             continue;
64         }
65         char client_id[50];
66         char client_name[50];
67         int registered = 0;
68         while (1)
69         {
70             char buffer[1024];
71             int bytes_read = read(client_socket, buffer, sizeof(buffer));
72             if (bytes_read == 0)
73             {
74                 close(client_socket);
75                 continue;
76             }
77             if (bytes_read < 0)
78             {
79                 perror("read");
80                 close(client_socket);
81                 continue;
82             }
83             if (strcmp(buffer, "register\n") == 0)
84             {
85                 strcpy(client_id, " ");
86                 strcpy(client_name, " ");
87                 registered = 0;
88             }
89             else if (strcmp(buffer, "quit\n") == 0)
90             {
91                 close(client_socket);
92                 continue;
93             }
94             else if (strcmp(buffer, "broadcast\n") == 0)
95             {
96                 broadcast_message(buffer, client_socket);
97             }
98             else if (registered)
99             {
100                 broadcast_message(buffer, client_socket);
101             }
102             else
103             {
104                 strcpy(client_id, buffer);
105                 strcpy(client_name, buffer);
106                 registered = 1;
107             }
108         }
109     }
110     return 0;
111 }
```

PROBLEMS OUTPUT TERMINAL PORTS

duc@Dell-615-5511:~/LTM/19_5/Bai_tap_4\$ nc 127.0.0.1 8888
Nhap theo cu phap: client_id: client_name
abc: Nguyen Van A
Dang ky thanh cong!
Xin chap
2026/05/25 10:46:59PM xyz:Tran Chao ban
[]

duc@Dell-615-5511:~/LTM/19_5/Bai_tap_4\$ nc 127.0.0.1 8888
Nhap theo cu phap: client_id: client_name
xyz: Tran B
Dang ky thanh cong!
2026/05/25 10:46:55PM abc:Nguyen Xin chap
Chao ban
[]

Ln 18, Col 20 Spaces: 4 UTF-8 LF Linux



```
23:02 25 Thg 5
File Edit Selection View Go Run Terminal Help
C chat_server.c U C telnet_server.c U F accounts.txt U X
home > duc > LTM > 19_5 > Bai_tap_4 > F accounts.txt
1 admin admin
2 guest nopass
3 test 123
```

PROBLEMS OUTPUT TERMINAL PORTS

duc@Dell-615-5511:~/LTM/19_5/Bai_tap_4\$ nc 127.0.0.1 8888
Nhap user va pass:
admin 123
Dang nhap that bai. Thu lai
Nhap user va pass:
admin admin
Dang nhap thanh cong

Nhap lenh: ls
accounts.txt
chat_server
chat_server.c
out.126877232527040.txt
telnet_server
telnet_server.c
Nhap lenh: []

Ln 3, Col 9 Spaces: 4 UTF-8 LF Plain Text

Bài 4.2 telnet_server

The image shows a Visual Studio Code editor window with a C program for a time server. The code is as follows:

```
150 listener =
151     socket(AF_INET,
152           SOCK_STREAM,
153           0);
154
155 int opt = 1;
156
157 setsockopt(listener,
158           SOL_SOCKET,
159           SO_REUSEADDR,
160           &opt,
161           sizeof(opt));
162
163 // Cấu hình server
164
165 server_addr.sin_family = AF_INET;
166
167 server_addr.sin_addr.s_addr =
168     INADDR_ANY;
169
170 server_addr.sin_port =
171     htons(PORT);
172
173 // Bind
174
175 bind(listener,
176     (struct sockaddr *)&server_addr,
```

The terminal window shows the following output:

```
duc@bell-615-5511:~/LTM/19_5/Bai_tap_4$ nc 127.0.0.1 8888
Nhập lệnh: GET TIME [format]
GET TIME dd/mm/yyyy
25/05/2026
GET TIME mm/dd/yy
05/25/26
Get Time avsd
Lệnh không hợp lệ
Get time dd/mm/yyyy
Lệnh không hợp lệ
```

Bài 4.3 time_server

Bài 4.4 http_server (prethead)

